

Algorithm and Dataset

Project Title: Tourism Analytics Dashboard

Objective

To analyze tourism data across different states of India and develop an interactive Power BI dashboard that provides insights into tourist arrivals, tourism revenue, and monthly trends.

Step-by-Step Procedure

Step 1: Collect the tourism dataset in CSV format.

Step 2: Import the dataset into Power BI Desktop.

Step 3: Perform data preprocessing:

- Check for missing values.
- Remove duplicate records.
- Verify data types.
- Standardize column names.
- Validate numerical values.

Step 4: Perform Exploratory Data Analysis (EDA):

- Analyze domestic and foreign tourist arrivals.
- Compare state-wise tourism performance.
- Study monthly tourism trends.
- Evaluate tourism revenue.

Step 5: Create DAX measures for key metrics:

- Total Domestic Tourists
- Total Foreign Tourists
- Total Tourists
- Total Tourism Revenue

Step 6: Design the Power BI dashboard using:

- KPI Cards
- Clustered Bar Chart
- Line Chart

- Pie/Donut Chart
- Table
- Slicers and Filters

Step 7: Validate the dashboard by checking calculations, chart accuracy, and filter functionality.

Step 8: Save, publish, and share the completed project.

Logic / Method Used

1. Collect tourism data.
2. Clean and validate the dataset.
3. Analyze tourism statistics.
4. Create interactive visualizations.
5. Generate business insights through the Power BI dashboard.

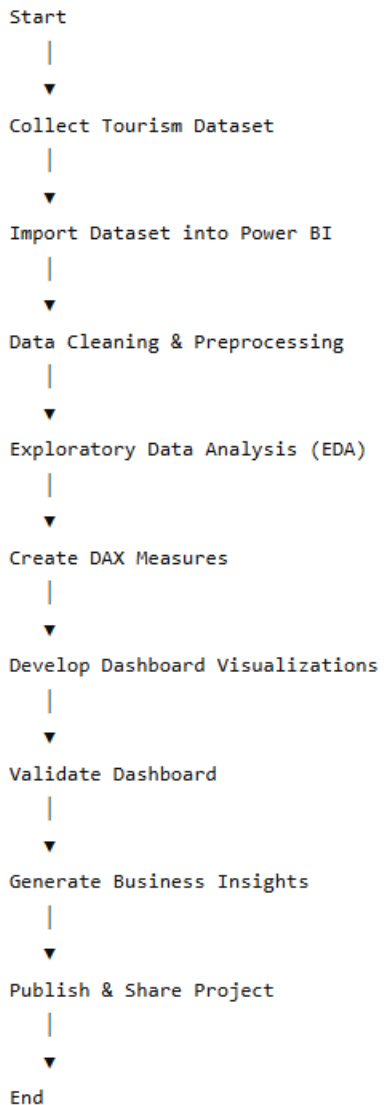
Model / Technique Applied

The project uses descriptive analytics and data visualization techniques rather than predictive modeling.

Techniques applied include:

- Data Cleaning
- Data Validation
- Exploratory Data Analysis (EDA)
- DAX Calculations
- Interactive Dashboard Design
- Business Intelligence Reporting

Process Flow



2. Dataset

Dataset Source

The tourism dataset was obtained from a tourism statistics dataset in **CSV format** (downloaded from Kaggle)

Number of Records (Samples)

- Total Records (Rows): **23**
- Total States/UTs: **36**

- Missing Values: **None**
- Duplicate Records: **None**

Features (Attributes/Columns)

Feature	Data Type	Description
State	Text	Name of the State/UT
Month	Text	Month of observation
Date	Date	Date of the record
Month_Number	Integer	Numeric representation of the month
Domestic Tourists	Number	Number of domestic visitors
Foreign Tourists	Number	Number of foreign visitors
Total Tourists	Number	Total number of tourists
Tourism Revenue	Decimal	Revenue generated through tourism

Total Features: 8

Training and Testing Data Split

Training Data: 80% - Testing Data: 20%

Data Format / Type

- File Format: CSV (.csv)
- Structured Data: Tabular
- Data Types:
 - Text
 - Date
 - Integer/Decimal/Numeri

Kumpati Vasanth
23H71A543

DVR & DR. HS MIC College of Technology